

# Data Modelling

## 1. Null percentage:

Days\_since\_prior\_order has a total of 206,209 null values which clearly indicates each user's first order. This null value has a percentage of 6.16% of total rows. These rows are not something which we should delete as it gives a clear indication of the very first order of each customer.

| null_val | null_percentage |
|----------|-----------------|
| 206209   | 6.16            |

## 2. Removing irrelevant column:

The orders table had a column "eval" which was generally an irrelevant column for our SQL report analysis. Hence, this is dropped from the table.

## 3. Item count mismatched:

After a careful analysis a very interesting fact found that there are 3 items under the products table which were never ordered as there are no entries for these 3 products under order\_products.

```
79 select
80 (select count(distinct product_id) from order_products) as item_cnt_order_products, /* 49685 */
81 (select count(distinct product_id) from products) as item_cnt_products,
82 (select count(distinct product_id) from products)-(select count(distinct product_id) from order_products) as item_diff, /* 49688 */
83
84
85 select product_id, product_name from products where product_id not in (select product_id from order_products); /* 3 never ordered items */
```

| item_cnt_order_products | item_cnt_products | item_diff |
|-------------------------|-------------------|-----------|
| 49685                   | 49688             | 3         |

  

| product_id | product_name                                    |
|------------|-------------------------------------------------|
| 3630       | Protein Granola Apple Crisp                     |
| 7045       | Unpeeled Apricot Halves in Heavy Syrup          |
| 46625      | Single Barrel Kentucky Straight Bourbon Whiskey |

## 4. Orphaned rows identified and removed:

75,000 rows existed in the orders table with no corresponding product data in order\_products.

### Investigation findings:

- All 75,000 orphaned orders belonged to customers who had other valid orders fully present in order\_products.
- Zero customers were entirely missing from order\_products.
- No customer would be lost from analysis after deletion.
- These rows represented empty order shells with no analytical value.

### Action taken:

- Backed up 75,000 rows to orders\_backup\_before\_cleanup.
- Deleted orphaned rows from the orders table.
- orders table reduced from 3421083 rows to 3346083 rows.

### Result:

- MIN order count per customer is now consistent at 3 across both the orders table and orders JOIN order\_products.

# Data Modelling

## 1. Adding missing items inside products table:

Some of the missing items were located under the table products prior to mapping the PK and FK. Certain product\_ids may cause issues when creating PKs and FKs since they are present under the table order\_products but not in the database products.

So, all the missing product\_ids from order\_products are added to products with the name "Unknown".

## 2. Primary Keys and Foreign Keys:

| Table          | Primary Key   | Foreign Key               |
|----------------|---------------|---------------------------|
| Orders         | order_id      | NA                        |
| Order_Products | NA            | order_id<br>product_id    |
| Products       | product_id    | department_id<br>aisle_id |
| Departments    | department_id | NA                        |
| Aisles         | aisle_id      | NA                        |

## 3. ERD (Entity Relationship Diagram):

